

TradingAgents V5 Benchmark — Comprehensive Results

Date: 2026-02-14 **Benchmark Version:** V5 (systematic/idiosyncratic risk framework) **Ticker:** AMBA (Ambarella Inc.) — "Hard" difficulty semiconductor **Period:** March – December 2025 (10 monthly snapshots) **Price Trajectory:** \$50.33 → \$70.84 (+40.7% over 10 months, bear-to-bull transition with late pullback)

Conclusion

No model generates consistent trading alpha on AMBA. The initial V5 benchmark (10 end-of-month dates) showed Sonnet 4.5 at 60% accuracy with +\$995 P&L — the first apparent alpha in any version. However, the tri-monthly validation (30 dates: 10th, 20th, and end-of-month × 10 months) reveals this was a statistical artifact:

- **Sonnet 30% accuracy at n=30** (down from 60% at n=10), P&L -\$875 (down from +\$995)
- **GPT 27% accuracy at n=30** (up from 20% at n=10), P&L +\$468 (up from -\$437)
- **Always-BUY: 47% at n=30** — both models underperform this trivial baseline at every horizon

Sonnet's 50% decision inconsistency on identical data (same cached inputs, different run) demonstrates that any "alpha" is inseparable from LLM non-determinism. The model essentially flips a coin on half its trades.

This represents a 5-version journey — each time apparent alpha was debunked by larger samples:

Version	Key Change	Best 1d Acc	vs Always-BUY	Alpha?	How Debunked
V1	Initial multi-agent system	45.8%*	-2.4pp	No*	Look-ahead bias, HOLD fallback bugs
V2	Fixed look-ahead + HOLD fallback	50.0% (n=4)	+1.8pp	Inconclusive	Too small (n=4)
V3	Expanded to 440 tasks	34.5%	-13.7pp	No	Broken data pipeline
V4	Fixed indicators + global news	40.0%	-10.0pp	No	Asymmetric SELL-biased prompt
V5 (EoM, n=10)	+SPY/sector context	60.0%	+10.0pp	Appeared so	Tri-monthly expansion to n=30
V5 (tri-monthly, n=30)	Same prompt, 3x dates	30.0%	-17.0pp	No	—

*V1's 45.8% was inflated by look-ahead bias (using live prices) and a HOLD fallback that masked parse failures. Multi-agent strategies showed 88-100% HOLD rates due to output parsing bugs. V2's 50% was

on only 4 data points (not statistically significant).

Final Results

Strategy Comparison (V5, AMBA, 10 dates)

Metric	Sonnet 4.5	GPT 5.1-codex-mini	Haiku 4.5
Next-Day Accuracy	60.0% (6/10)	20.0% (2/10)	30.0% (3/10)
5-Day Accuracy	20.0% (2/10)	60.0% (6/10)	40.0% (4/10)
Total P&L	+\$995.21	-\$437.42	+\$331.44
Max Drawdown	\$375.49	\$723.51	\$9.13
Avg Confidence	0.64	0.41	0.64
BUY %	30%	40%	10%
SELL %	30%	20%	10%
HOLD %	40%	40%	80%
Overall Horizon Acc	31.3%	34.3%	14.9%
Est. Cost	~\$1.00	~\$0.80	~\$0.20

Comparison with V2 Benchmark (AMBA Jan-Apr 2025, 4 dates)

Metric	V2 Opus 4.5	V2 OpenAI Single	V5 Sonnet 4.5
Next-Day Accuracy	50.0%	25.0%	60.0%
Total P&L	+\$696.25	+\$645.82	+\$995.21
Avg Confidence	0.69	0.33	0.64
Decision Mix	50% SELL, 50% HOLD	75% SELL, 25% HOLD	30% BUY, 30% SELL, 40% HOLD
Data Quality	No indicators, no market context	No indicators, no market context	Full: indicators + SPY + sector
Market Period	Bear (−37.5%)	Bear (−37.5%)	Mixed (+40.7%)

Key improvement over V2: V5 Sonnet produces **balanced BUY/SELL/HOLD decisions** (30/30/40) vs V2's SELL-only bias (50/75% SELL). V2 strategies never issued a single BUY across all 16 decisions. V5 Sonnet issued 3 BUYs, 2 of which were profitable.

Decision-by-Decision (All 3 Models)

Date	Close	Next-Day	Regime	Sonnet 4.5	GPT 5.1-codex-mini	Haiku 4.5
Mar 31	\$50.33	+0.26%	Bear	BUY (0.55)	SELL (0.35)	HOLD (0.65)
Apr 30	\$47.99	+0.35%	Bear	HOLD (0.70)	HOLD (0.36)	HOLD (0.55)
May 31	\$52.64	+0.99%	Sideways	SELL (0.70)	SELL (0.52)	HOLD (0.72)
Jun 30	\$66.07	-2.47%	Bull	BUY (0.72)	BUY (0.34)	HOLD (0.60)
Jul 31	\$66.09	-3.56%	Sideways	SELL (0.70)	HOLD (0.42)	SELL (0.68)
Aug 31	\$82.48	-2.92%	Bull	HOLD (0.45)	BUY (0.52)	HOLD (0.72)
Sep 30	\$82.52	+3.10%	Sideways	HOLD (0.70)	BUY (0.61)	HOLD (0.68)
Oct 31	\$85.23	+0.06%	Sideways	HOLD (0.70)	BUY (0.30)	BUY (0.62)
Nov 30	\$74.18	-4.39%	Bear	SELL (0.75)	HOLD (0.32)	HOLD (0.55)
Dec 31	\$70.84	+6.10%	Sideways	BUY (0.40)	HOLD (0.38)	HOLD (0.65)

Bold = pivotal decisions (discussed in detail below).

Correct decisions highlighted:

- **Sonnet:** Mar ✓, Apr ✓, May X, Jun X, Jul ✓, Aug X, Sep X, Oct ✓, Nov ✓, Dec ✓ → **6/10 (60%)**
- **GPT:** Mar X, Apr ✓, May X, Jun X, Jul X, Aug X, Sep ✓, Oct X, Nov X, Dec X → **2/10 (20%)**
- **Haiku:** Mar ✓, Apr ✓, May X, Jun X, Jul ✓, Aug X, Sep X, Oct X, Nov X, Dec X → **3/10 (30%)**

P&L by Trade

Date	Sonnet 4.5	GPT 5.1-codex-mini	Haiku 4.5
Mar 31	+\$10.83	-\$40.83	\$0.00
Apr 30	\$0.00	\$0.00	\$0.00
May 31	-\$113.78	-\$113.78	\$0.00
Jun 30	-\$261.71	-\$261.71	\$0.00
Jul 31	+\$340.57	\$0.00	+\$340.57
Aug 31	\$0.00	-\$307.19	\$0.00
Sep 30	\$0.00	+\$295.23	\$0.00
Oct 31	\$0.00	-\$9.13	-\$9.13
Nov 30	+\$424.47	\$0.00	\$0.00
Dec 31	+\$594.83	\$0.00	\$0.00
Total	+\$995.21	-\$437.42	+\$331.44

Sonnet's P&L is driven by 3 high-conviction directional calls: Jul SELL (+\$341), Nov SELL (+\$424), Dec BUY (+\$595). Its losses were smaller (-\$114, -\$262). Net: +\$995.

V1 → V2 → V3 → V4 → V5 Evolution

V1: Multi-Agent System with Hidden Bugs (503 tasks)

The V1 benchmark (2026-01-28) tested the full multi-agent architecture: 503 tasks, 6 tickers, 7 strategies, 12 months. Surface-level results looked promising — single_5.1-codex-mini hit 45.8% accuracy with +\$1,173 P&L. But two critical bugs invalidated the results:

1. **Look-ahead bias:** `yf.download()` fetched live current prices via `get_current_quote()`, meaning the model could "see" future price data when making decisions.
2. **HOLD fallback:** When the LLM output couldn't be parsed, the system silently defaulted to HOLD instead of raising an error. This meant multi-agent strategies (which had complex output formats prone to parse failures) showed 88-100% HOLD rates — they weren't choosing HOLD, the parser was failing.

Strategy Type	HOLD Rate	Why
Multi-agent (research_manager, etc.)	88-100%	Output parsing failures → silent HOLD fallback
Single-agent (single_5.1-codex-mini)	47%	Simpler output format, fewer parse failures

AMBA-specific V1 results: Best accuracy was 42% (ma_5.2+5.1-codex-mini and ma_4o strategies), but these were inflated by look-ahead bias.

V2: Methodology Fix (16 tasks)

The V2 benchmark (2026-02-02) fixed both bugs:

- Replaced live `yf.download()` with historical OHLCV `decision_date_close`
- Removed HOLD fallback — parse failures now raise errors
- Added content normalization for consistent LLM parsing

Results on 4 dates × 4 strategies (16 tasks total):

Strategy	1d Accuracy	P&L	HOLD %
Opus Single	50.0%	+\$696	50%
OpenAI Single	25.0%	+\$646	0%
Multi-agent (ma_Opus)	0.0%	\$0	100%
Multi-agent (ma_OpenAI)	0.0%	\$0	100%

Multi-agent strategies were still 100% HOLD even after fixing the fallback — confirming the architecture itself couldn't produce actionable decisions. V2 proved single-agent was the path forward, but n=4 was too small for statistical significance.

The Problem: No Alpha in V3 (440 tasks)

The V3 benchmark (2026-02-11) was the definitive large-scale test: 440 tasks, 4 models, 10 tickers, 11 months. The result was unambiguous — **no model generated trading alpha**.

Model	1d Accuracy	vs Always-BUY	P&L	\$/task
Haiku 4.5	32.7%	-15.5pp	-\$86	\$0.02
gpt-4o-mini	31.8%	-16.4pp	-\$1,096	\$0.03
gpt-5.1-codex-mini	31.8%	-16.4pp	-\$37	\$0.08
Sonnet 4.5	34.5%	-13.7pp	+\$1,280	\$0.10
Always-BUY	48.2%	—	—	\$0.00

All models clustered at ~33% accuracy with completely overlapping confidence intervals. Spending 5x more on Sonnet (\$0.10) vs Haiku (\$0.02) bought zero measurable accuracy improvement.

Root cause: The data pipeline was broken.

V3 Data Pipeline Failures

Data Category	V3 Status	Impact
Technical Indicators (RSI, MACD, SMA)	BROKEN — date string passed as indicator name	Models made decisions without any technical analysis
Global/Macro News	BROKEN — date string passed where integer expected	No systematic risk context
Social Sentiment	EMPTY — all providers blocked or missing keys	No crowd sentiment data
Market OHLCV	Working	Price history available
Fundamentals (P/E, EPS, etc.)	Working	Balance sheet, income, cash flow available
Ticker-Specific News	Working	Company news available

LLMs were essentially making trading decisions with raw price data, fundamentals, and company news — but **without the technical signals traders rely on most** (RSI, MACD, Bollinger Bands, SMA) and **without any market-wide context**.

V4: Fixed Data Pipeline (2026-02-13)

Three bugs were fixed:

Bug	V3 Behavior	V4 Fix
Indicators	<code>get_indicators("AMBA", "2025-06-30", ...)</code> — date as indicator name	Fixed parameter ordering; added <code>_normalize_indicator()</code> alias map
Global News	<code>get_global_news("2025-06-30")</code> — string where int expected	Fixed call signature in <code>_fetch_data_safe()</code> schema introspection

Bug	V3 Behavior	V4 Fix
yfinance Look-ahead	yf.download() used live current price for get_current_quote()	Replaced with historical OHLCV decision_date_close

Data quality improvement (verified on AMBA 2025-06-30):

Category	V3 (broken)	V4 (fixed)	Change
Technical Indicators	82 chars (error message)	6,628 chars (RSI/MACD/SMA data)	80x more data
Global News	288 chars (parse error)	69,138 chars (real articles)	240x more data

V4 smoke test results (10 tickers, single date 2025-06-30, Haiku):

Metric	V3 Haiku (broken)	V4 Haiku (fixed)	Change
1d Accuracy	30%	40%	+10pp
1w Accuracy	20%	30%	+10pp
BUY %	10%	0%	-10pp
SELL %	20%	40%	+20pp
HOLD %	70%	60%	-10pp

Accuracy improved at short horizons (+10pp at 1d and 1w), confirming the data quality fix had real impact. **But a new problem emerged: systematic bearish bias.** V4 produced 0% BUY decisions — the model never once said BUY.

V4 Root Cause: Asymmetric Decision Rules

Analysis of the V4 prompt revealed the cause:

Rule	Direction	How Often Triggered
Price < SMA50 AND SMA200	SELL/HOLD	Frequently (catch-all)
RSI > 70 + declining	SELL/HOLD	Sometimes (NVDA, etc.)
MACD bearish crossover	SELL	Sometimes
Sentiment bullish + price bearish	Prioritize price (= SELL)	Always
RSI < 30	BUY	Almost never (all RSI 47-77)

The prompt had **4 SELL/HOLD triggers** and only **1 BUY trigger** (RSI < 30), which almost never fires in a bull market. The data pipeline was fixed, but the decision framework was structurally biased.

V5: Systematic/Idiosyncratic Risk Framework (2026-02-14)

V5 addressed both the bearish bias and the lack of market context with three changes:

1. Added SPY Broad Market Context (Systematic Risk)

The model now sees 90 days of SPY OHLCV and 6 technical indicators (RSI, MACD, SMA50, SMA200, Bollinger, ATR) alongside the ticker data. This provides market regime detection:

- SPY above 50+200 SMA with positive MACD = **BULL** regime
- SPY below both with negative MACD = **BEAR** regime
- Mixed = **NEUTRAL** regime

2. Added Sector ETF Context (Industry Risk)

Each ticker is mapped to its SPDR sector ETF (AMBA→SMH, AAPL→XLK, JPM→XLF, etc.). The model sees if the sector is leading, lagging, or in line with the broad market.

3. Replaced Asymmetric Rules with Balanced Decision Matrix

Regime	Bullish Ticker	Bearish Ticker
BULL	Strong BUY	HOLD (support limits downside)
BEAR	HOLD (headwinds limit upside)	Strong SELL
NEUTRAL	Follow ticker signals	Follow ticker signals

5 balanced bullish signals (price > SMA50, RSI 40-60 rising, MACD positive, positive earnings, strong FCF) and 5 balanced bearish signals replaced the asymmetric rule set.

4. Removed Dead Social Sentiment Section

The social sentiment section always showed error messages, wasting context tokens and potentially biasing the model.

Key Findings

1. The V5 framework works — the HOLD bias is model-specific, not prompt-level

This was the critical diagnostic question: does the V5 prompt produce balanced decisions? Yes — but only with capable models.

Model	BUY %	SELL %	HOLD %	Directional %
Haiku 4.5	10%	10%	80%	20%
Sonnet 4.5	30%	30%	40%	60%
GPT 5.1-codex-mini	40%	20%	40%	60%

Sonnet and GPT both dropped HOLD from 80% to 40%. When tested directly, Haiku's reasoning correctly identified market regimes and applied the Step 3 combine matrix — but it hedged at the final decision, defaulting to HOLD when signals were mixed. This is a model capability limitation, not a prompt failure.

2. Sonnet 4.5 achieves genuine short-term edge

Sonnet's 60% next-day accuracy on AMBA (n=10) is above the 50% Always-BUY baseline for the first time in any benchmark version. While n=10 is too small for statistical significance (Bootstrap 95% CI: [30%, 90%]), the pattern is consistent:

- **High-confidence directional calls are correct:** Jul SELL (0.70) → +\$341, Nov SELL (0.75) → +\$424, Dec BUY (0.40) → +\$595
- **Walk-forward improves:** P1 accuracy 50.0% → P2 accuracy 75.0% (+25pp), suggesting generalization not overfitting
- **Sonnet's SELL signal is real:** 66.7% accuracy at 1d and 3d horizons for SELL decisions

3. GPT 5.1-codex-mini has medium-term advantage but poor short-term precision

Horizon	Sonnet 4.5	GPT 5.1-codex-mini	Winner
1d	50.0%	10.0%	Sonnet
3d	40.0%	40.0%	Tie
1w	20.0%	60.0%	GPT
2w	20.0%	40.0%	GPT
4w	30.0%	40.0%	GPT
8w	33.3%	33.3%	Tie
13w	25.0%	12.5%	Sonnet

GPT wins at 1w-4w horizons but with very low confidence (avg 0.41 — nearly all decisions are "uncertain"). Sonnet wins at the extremes (1d, 13w) with well-calibrated confidence. The models have complementary strengths.

4. Confidence calibration separates winners from losers

Model	Avg Conf	Range	High-Conf Accuracy	Low-Conf Accuracy
Sonnet 4.5	0.64	0.40–0.75	57% (7 trades)	67% (3 trades)
GPT 5.1-codex-mini	0.41	0.30–0.61	100% (1 trade)	11% (9 trades)
Haiku 4.5	0.64	0.55–0.72	25% (8 trades)	50% (2 trades)

Sonnet expresses meaningful variation in confidence (0.40–0.75). GPT barely varies (0.30–0.61) and is almost always below the 0.5 threshold. Haiku also has narrow range but high absolute values (0.55–0.72), creating false confidence.

5. Data pipeline fixes alone are insufficient — framework matters

The V3→V4 transition showed that fixing broken data improved accuracy by ~10pp (30% → 40% at 1d). But V4 still had no alpha due to the bearish bias in the decision rules. The V4→V5 transition — changing the prompt framework without changing the data — was the bigger lever for Sonnet (+20pp from V4's 40% to V5's 60%). **The right framework on the right model matters more than data quality alone.**

6. HOLD remains structurally wrong for volatile stocks

HOLD accuracy by model across all horizons:

Model	HOLD 1d	HOLD 1w	HOLD 4w	HOLD 13w
Sonnet 4.5	25.0%	0.0%	25.0%	0.0%
GPT 5.1-codex-mini	0.0%	25.0%	0.0%	25.0%
Haiku 4.5	12.5%	12.5%	0.0%	0.0%

AMBA's average daily move exceeds the HOLD threshold most of the time. Models that default to HOLD (Haiku at 80%) are structurally penalized. The optimal strategy for volatile stocks is to take directional positions, even with imperfect accuracy.

7. Progression from V1 to V5 shows compounding improvements

Capability	V1 (Jan 28)	V2 (Feb 2)	V3 (Feb 11)	V4 (Feb 13)	V5 (Feb 14)
Look-ahead bias	Present	Fixed	Fixed	Fixed	Fixed
HOLD fallback on errors	Present	Removed	Removed	Removed	Removed
Technical indicators	Available*	Missing	Broken	Fixed	Fixed
Global news	Available*	Missing	Broken	Fixed	Fixed
Market regime (SPY)	None	None	None	None	Added
Sector context (ETF)	None	None	None	None	Added
Decision framework	Multi-agent	Multi-agent	Basic	Basic (asymmetric)	Balanced (systematic/idiosyncratic)
Architecture	Multi-agent (7 strategies)	Multi-agent + single	Single-agent only	Single-agent only	Single-agent only
Best 1d accuracy	45.8%** (n=72)	50.0% (Opus, n=4)	34.5% (Sonnet, n=110)	40.0% (Haiku, n=10)	60%→ 30% (Sonnet, n=10→30)

Capability	V1 (Jan 28)	V2 (Feb 2)	V3 (Feb 11)	V4 (Feb 13)	V5 (Feb 14)
Best P&L	+\$1,173**	+\$696 (Opus)	+\$1,280 (Sonnet)	N/A	+\$995→**-\$875** (Sonnet, n=10→30)
BUY decisions issued	0-12%***	0/16	5-19%	0%	10-30% (date-dependent)
Multi-agent HOLD rate	88-100%	100%	N/A	N/A	N/A

*V1 data pipeline was functional but results were inflated by look-ahead bias (live prices) and HOLD fallback masking errors. **V1 metrics are not directly comparable due to look-ahead bias. ***V1 multi-agent strategies issued 0% BUY; only single-agent strategies produced BUY decisions.

Pivotal Decisions

June 30 — The Bull Market Test

AMBA at \$66.07 after a massive +30% rally from the April bottom. SPY indicators showed BULL regime. The key question: is this a top or continuation?

- **Sonnet** called **BUY (0.72)** — highest confidence of the date. Wrong next-day (-2.47%), but correct at 3d through 13w. Horizon score: 6/7.
- **GPT** also called **BUY (0.34)** — same direction but very low confidence. Also 6/7 on horizons.
- **Haiku** said **HOLD (0.60)** — missed the opportunity entirely. Only 1/7 on horizons.

This date illustrates the Haiku HOLD problem: the regime was clearly bullish, the framework says "BULL + bullish ticker = strong BUY," but Haiku hedged. Sonnet and GPT followed the framework correctly.

July 31 — Sonnet's Best SELL

AMBA at \$66.09, essentially flat from June. Market regime shifted to sideways. Next day crashed -3.56%.

- **Sonnet** called **SELL (0.70)** — correct, +\$341 P&L. Also correct at 1d, 3d, and 1w horizons (3/7).
- **Haiku** also called **SELL (0.68)** — one of only two directional calls Haiku made. Also +\$341.
- **GPT** said **HOLD (0.42)** — missed it. 0/7 on horizons.

Both Anthropic models detected the exhaustion signal. GPT's low confidence reflects genuine uncertainty but missed the trade.

November 30 — The High-Conviction SELL

AMBA had peaked at \$90.97 in late November and pulled back to \$74.18 (-18% from peak). Market regime was BEAR. Next day dropped another -4.39%.

- **Sonnet** called **SELL (0.75)** — the highest confidence of any call in the entire benchmark. Correct: +\$424 P&L. 4/6 horizons correct.

- **GPT** said **HOLD (0.32)** — extremely low confidence, effectively uncertain. Missed the trade.
- **Haiku** said **HOLD (0.55)** — defaulted to inaction despite bear regime.

This was Sonnet's signature trade: high conviction, correct direction, large P&L. The 0.75 confidence was meaningfully above its average (0.64), suggesting the model recognized this as a stronger signal than usual.

December 31 — The Contrarian BUY

AMBA at \$70.84, down 22% from the October peak. The sentiment was bearish but the stock was oversold. Next day surged +6.10% — the largest single-day move in the dataset.

- **Sonnet** called **BUY (0.40)** — its lowest confidence BUY, reflecting uncertainty. But correct: +\$595, the single most profitable trade.
- **GPT** said **HOLD (0.38)** — almost identical low confidence, but chose inaction.
- **Haiku** said **HOLD (0.65)** — high confidence HOLD, wrong.

Sonnet's willingness to act on a low-conviction BUY when the framework indicated a potential opportunity separated it from the other models. GPT had nearly the same read (0.38 confidence) but defaulted to HOLD.

Multi-Horizon Analysis

Horizon Comparison (V5, All Models + Baselines)

Horizon	Sonnet 4.5	GPT 5.1-codex-mini	Haiku 4.5	Always-BUY	Momentum-5d	Random
1d	50.0%	10.0%	10.0%	50.0%	30.0%	33.3%
3d	40.0%	40.0%	20.0%	60.0%	50.0%	33.3%
1w	20.0%	60.0%	40.0%	50.0%	60.0%	33.3%
2w	20.0%	40.0%	20.0%	50.0%	60.0%	33.3%
4w	30.0%	40.0%	10.0%	60.0%	70.0%	33.3%
8w	33.3%	33.3%	0.0%	77.8%	55.6%	33.3%
13w	25.0%	12.5%	0.0%	62.5%	50.0%	33.3%

Key observation: GPT 5.1-codex-mini matches or beats Momentum-5d at 1w (60% vs 60%) and approaches it at 4w (40% vs 70%). This is the first time an LLM has been competitive with a simple momentum baseline at medium-term horizons.

BUY Decision Accuracy by Horizon

Horizon	Sonnet 4.5	GPT 5.1-codex-mini
1d	66.7%	25.0%

Horizon	Sonnet 4.5	GPT 5.1-codex-mini
3d	66.7%	75.0%
1w	33.3%	75.0%
2w	33.3%	50.0%
4w	33.3%	75.0%
8w	100.0%	75.0%
13w	100.0%	25.0%

Sonnet's BUY calls are remarkably durable — 100% correct at 8w and 13w. GPT's BUY calls are better at medium-term (1w-4w at 50-75%) but deteriorate at longer horizons.

SELL Decision Accuracy by Horizon

Horizon	Sonnet 4.5	GPT 5.1-codex-mini
1d	66.7%	0.0%
3d	66.7%	50.0%
1w	33.3%	50.0%
2w	33.3%	100.0%
4w	33.3%	50.0%
8w	33.3%	0.0%
13w	0.0%	0.0%

Sonnet has a genuine 1d SELL signal (66.7% accuracy). This is consistent with the V3 finding where Sonnet was the only model with a real SELL signal (68.2% at 1d, n=22 across 10 tickers).

Cross-Version SELL Signal Analysis

Sonnet 4.5's SELL signal has been consistent across benchmark versions:

Version	SELL 1d Accuracy	n	Statistical?
V3 (440 tasks, 10 tickers)	68.2%	22	Yes (vs 46% base)
V5 (10 tasks, AMBA)	66.7%	3	Consistent pattern

No other model shows this consistency. GPT's SELL accuracy is 0-57% depending on version. Haiku's is 41%. Sonnet's SELL signal appears to be a genuine capability, not noise.

Decision Agreement Matrix

How often each pair of V5 models agreed on the same decision:

	Sonnet 4.5	GPT 5.1-codex-mini	Haiku 4.5
Sonnet 4.5	—	4/10 (40%)	3/10 (30%)
GPT 5.1-codex-mini	4/10	—	3/10 (30%)
Haiku 4.5	3/10	3/10	—

Low agreement across the board (30-40%). The models disagree more than they agree, suggesting they are processing the same data very differently. Haiku's 80% HOLD strategy naturally disagrees with the more directional models.

Dates where all 3 models agreed:

- **Apr 30:** All three said HOLD — correct (market moved <0.35%)
- No other unanimous agreement.

Confidence Analysis

Sonnet 4.5: Confidence vs Outcome

Date	Decision	Conf	1d Correct?	P&L	Horizons
Nov 30	SELL	0.75	Y	+\$424	4/6
Jun 30	BUY	0.72	N	-\$262	6/7
May 31	SELL	0.70	N	-\$114	1/7
Jul 31	SELL	0.70	Y	+\$341	3/7
Apr 30	HOLD	0.70	Y	\$0	0/7
Sep 30	HOLD	0.70	N	\$0	0/7
Oct 31	HOLD	0.70	Y	\$0	1/7
Mar 31	BUY	0.55	Y	+\$11	3/7
Aug 31	HOLD	0.45	N	\$0	1/7
Dec 31	BUY	0.40	Y	+\$595	2/5

Observations:

- Sonnet's high-confidence directional calls (>0.65) are 2/3 correct at 1d but have strong multi-horizon performance (Jun BUY: 6/7, Nov SELL: 4/6)
- The 0.70 HOLD cluster suggests Sonnet maps "insufficient evidence for direction" to exactly 0.70 confidence — a systematic anchor
- The most profitable trade (Dec BUY, +\$595) was the lowest-confidence BUY (0.40) — willingness to act on low conviction paid off

GPT 5.1-codex-mini: Confidence Problem

GPT's average confidence is 0.41 — meaning 9 of 10 trades are classified as "low confidence." Its single high-confidence trade (Sep BUY, 0.61) was correct. The model appears to lack the capability to express conviction, making its confidence signal nearly useless for position sizing.

Walk-Forward Validation

Split date: 2025-09-01 (Period 1: Mar-Aug, 6 dates | Period 2: Sep-Dec, 4 dates)

Strategy	Period 1 (in-sample)	Period 2 (out-of-sample)	Delta
Sonnet 4.5	3/6 (50.0%)	3/4 (75.0%)	+25.0%
GPT 5.1-codex-mini	1/6 (16.7%)	1/4 (25.0%)	+8.3%
Haiku 4.5	3/6 (50.0%)	0/4 (0.0%)	-50.0%

Sonnet improves out-of-sample — from 50% to 75% accuracy. This is the opposite of overfitting and suggests the model generalizes well. Haiku collapses from 50% to 0% out-of-sample, confirming its decisions are noise rather than signal.

Cost Efficiency Analysis

Model	\$/task	Tasks	Total Cost	1d Acc	P&L	P&L/\$ Spent
Haiku 4.5	\$0.02	10	~\$0.20	30%	+\$331	\$1,655
GPT 5.1-codex-mini	\$0.08	10	~\$0.80	20%	-\$437	-\$546
Sonnet 4.5	\$0.10	10	~\$1.00	60%	+\$995	\$995

Haiku has the best P&L-per-dollar due to its ultra-low cost, but its accuracy is poor (30%) and its out-of-sample performance collapsed. Sonnet is the only model with both positive P&L and positive out-of-sample delta.

Tri-Monthly Deep Analysis (n=30 per model)

The initial V5 benchmark tested 10 end-of-month dates only. To determine whether Sonnet's 60% accuracy was genuine or date-dependent luck, we ran the same prompt on 30 dates: the 10th, 20th, and end-of-month for each month, March–December 2025.

Run ID: v2_20260214_193006 | **Tasks:** 60 (30 dates × 2 models) | **Failed:** 0

The Answer: No Durable Alpha

With 3x the data, the picture reverses entirely:

Metric	End-of-Month Only (n=10)	All 30 Dates (n=30)	Change
Sonnet 1d accuracy	60%	30%	-30pp

Metric	End-of-Month Only (n=10)	All 30 Dates (n=30)	Change
Sonnet P&L	+\$995	-\$875	-\$1,870
GPT 1d accuracy	20%	27%	+7pp
GPT P&L	-\$437	+\$468	+\$905
Always-BUY 1d	50%	47%	-3pp

Sonnet's 60% was a statistical artifact of the end-of-month sample. When tested across all three date windows, accuracy drops to 30% — below random (33%). GPT's apparent underperformance at end-of-month (10%) was equally misleading; its full-sample accuracy is 27%, roughly at random.

Neither model beats Always-BUY at any date group. The clearest finding: buying AMBA on any date and holding outperforms both LLMs across almost every horizon and date group.

Accuracy by Date Group and Horizon

Sonnet 4.5:

Horizon	10th (n=10)	20th (n=10)	EoM (n=10)	All (n=30)	Spread
1d	3/10 (30%)	2/10 (20%)	4/10 (40%)	9/30 (30%)	20pp
3d	1/10 (10%)	2/10 (20%)	4/10 (40%)	7/30 (23%)	30pp
1w	3/10 (30%)	3/10 (30%)	4/10 (40%)	10/30 (33%)	10pp
2w	1/10 (10%)	1/10 (10%)	3/10 (30%)	5/30 (17%)	20pp
4w	1/10 (10%)	4/10 (40%)	2/10 (20%)	7/30 (23%)	30pp
8w	2/10 (20%)	1/9 (11%)	2/9 (22%)	5/28 (18%)	11pp
13w	3/9 (33%)	0/8 (0%)	2/8 (25%)	5/25 (20%)	33pp

GPT 5.1-codex-mini:

Horizon	10th (n=10)	20th (n=10)	EoM (n=10)	All (n=30)	Spread
1d	2/10 (20%)	5/10 (50%)	1/10 (10%)	8/30 (27%)	40pp
3d	1/10 (10%)	3/10 (30%)	4/10 (40%)	8/30 (27%)	30pp
1w	2/10 (20%)	5/10 (50%)	6/10 (60%)	13/30 (43%)	40pp
2w	4/10 (40%)	5/10 (50%)	3/10 (30%)	12/30 (40%)	20pp
4w	4/10 (40%)	3/10 (30%)	4/10 (40%)	11/30 (37%)	10pp
8w	4/10 (40%)	5/9 (56%)	3/9 (33%)	12/28 (43%)	22pp
13w	3/9 (33%)	3/8 (38%)	1/8 (12%)	7/25 (28%)	25pp

Key observation: Both models show 20-40pp spreads between date groups at every horizon. This means accuracy is not stable — it fluctuates wildly depending on which day of the month the model is

asked. This is the signature of noise, not signal.

Decision Distribution: Sonnet's SELL Bias Returns

Model	Date Group	BUY	SELL	HOLD
Sonnet	10th	10%	40%	50%
Sonnet	20th	0%	50%	50%
Sonnet	EoM	20%	30%	50%
Sonnet	ALL	10%	40%	50%
GPT	10th	50%	10%	40%
GPT	20th	50%	30%	20%
GPT	EoM	40%	20%	40%
GPT	ALL	47%	20%	33%

Sonnet has a persistent **SELL bias** (40% SELL vs 10% BUY overall). On the 20th of each month, it never once called BUY. The "balanced" 30/30/40 distribution from the end-of-month run was not representative — it was the most balanced of the three date groups.

GPT maintains a consistent BUY-heavy distribution (~47% BUY) across all date groups.

P&L by Date Group

Model	10th	20th	EoM	Total
Sonnet	-\$1,293	-\$119	+\$538	-\$875
GPT	+\$277	+\$565	-\$374	+\$468

Sonnet's P&L is entirely dependent on date selection. End-of-month is the only profitable date group (+\$538), while 10th-of-month loses \$1,293. The Oct 10 SELL call at 0.78 confidence (the highest confidence trade in the entire dataset) lost \$1,122 alone — proving that even maximum confidence doesn't predict accuracy.

Holding Power Analysis

This is the critical test: if a model gets the 1d direction right, does the call "hold" at longer horizons?

GPT — If correct at 1d, still correct at:

3d	1w	2w	4w	8w	13w
5/8 (62%)	4/8 (50%)	4/8 (50%)	4/8 (50%)	5/8 (62%)	3/7 (43%)

Sonnet — If correct at 1d, still correct at:

3d	1w	2w	4w	8w	13w
5/9 (56%)	2/9 (22%)	1/9 (11%)	1/9 (11%)	1/9 (11%)	2/8 (25%)

Sonnet has no holding power. When it gets the 1d call right, only 11% of those calls are still correct at 2w or 4w. Its correct 1d predictions rapidly reverse. By contrast, GPT's correct 1d calls maintain 50% accuracy out to 4 weeks — not great, but meaningfully better than Sonnet.

If **WRONG** at 1d, how often correct later?

- **GPT** wrong at 1d (n=22): eventually correct at 1w in 41% of cases
- **Sonnet** wrong at 1d (n=21): eventually correct at 1w in 38% of cases

Both models show similar "recovery" rates, suggesting the longer-horizon accuracy is more a function of AMBA's price trajectory than model intelligence.

HOLD Is Almost Always Wrong

For trades where each model called HOLD, what would have been the right call?

Sonnet's 15 HOLD trades:

Horizon	BUY would win	SELL would win	HOLD correct	HOLD wrong
1d	47%	40%	13%	87%
1w	33%	47%	20%	80%
4w	13%	67%	20%	80%
13w	67%	33%	0%	100%

HOLD is correct only 0–20% of the time. AMBA is too volatile for HOLD to be a rational decision at any horizon. Sonnet's 50% HOLD rate structurally costs it P&L — every HOLD is a missed directional opportunity.

Directional Trades Only: Sonnet's Edge Appears

When filtering to **directional trades only** (excluding HOLDs):

Model	n	1d	3d	1w	2w	4w
Sonnet	15	46.7%	46.7%	46.7%	33.3%	26.7%
GPT	20	35.0%	40.0%	55.0%	60.0%	55.0%

When Sonnet actually commits to a direction, it's 46.7% accurate at 1d — close to random but better than GPT's 35%. However, GPT dominates 1w–4w with 55–60% accuracy on directional trades.

LLM Consistency: Same Data, Different Decisions

Comparing end-of-month decisions between this run and the previous V5 run (identical cached data):

Model	Same Decision	Different	Consistency
GPT 5.1-codex-mini	8/10	2/10	80%
Sonnet 4.5	5/10	5/10	50%

Sonnet is essentially flipping a coin on half its EoM dates. Given the exact same market data, Sonnet changed its decision on 5/10 dates (Mar, Aug, Sep, Nov, Dec). This non-determinism directly explains how the same model can show 60% accuracy on one run and 40% on another.

GPT is more consistent (80%) — but its consistent decisions are often wrong, so consistency alone doesn't create alpha.

Always-BUY Comparison by Date Group

Date Group	Horizon	Always-BUY	Sonnet	GPT	Sonnet vs BUY	GPT vs BUY
10th	1d	40%	30%	20%	-10pp	-20pp
10th	13w	67%	33%	33%	-33pp	-33pp
20th	1d	50%	20%	50%	-30pp	+0pp
20th	13w	75%	0%	38%	-75pp	-38pp
EoM	1d	50%	40%	10%	-10pp	-40pp
EoM	13w	62%	25%	12%	-38pp	-50pp

No model beats Always-BUY in any date group at any horizon. The closest is GPT matching Always-BUY at 20th/1d (50% vs 50%). Sonnet at 20th/13w is 0% accurate — 75pp behind Always-BUY.

Confidence Calibration (n=30)

Sonnet:

Confidence	n	1d Accuracy	Avg P&L
Medium (0.4-0.6)	3	1/3 (33%)	+\$62
High (0.6-0.75)	23	7/23 (30%)	+\$7
Very High (≥ 0.75)	4	1/4 (25%)	-\$307

Sonnet's highest-confidence bucket (≥ 0.75) has the **worst accuracy (25%) and worst P&L (-\$307)**. Confidence is inversely correlated with accuracy. The Oct 10 SELL at 0.78 (losing \$1,122) is the clearest example: maximum confidence, maximum loss.

GPT:

Confidence	n	1d Accuracy	Avg P&L
Low (< 0.4)	8	2/8 (25%)	+\$58
Medium (0.4-0.6)	17	5/17 (29%)	+\$20

Confidence	n	1d Accuracy	Avg P&L
High (0.6-0.75)	3	1/3 (33%)	-\$115

GPT's confidence signal is flat — all buckets are 25-33%. No filtering possible.

Verdict

The V5 prompt framework does not generate consistent trading alpha on AMBA. This conclusion holds across all 30 data points, 7 horizons, and both models:

1. **Sonnet's 60% accuracy was a 10-point fluke.** The same model on the same EoM dates only reproduces 50% of its original decisions. On 30 dates, accuracy drops to 30%. P&L goes from +\$995 to -\$875.
2. **Neither model beats Always-BUY.** At n=30, both models underperform the simplest possible baseline across all date groups and almost all horizons.
3. **Accuracy is highly date-dependent** (20-40pp spreads between date groups). This proves the models are not responding to stable market features — they are producing pseudo-random outputs that happen to align with market moves on some dates but not others.
4. **Sonnet has zero holding power.** Only 11% of correct 1d calls remain correct at 2w+. This means even when the model is "right," it's right for the wrong reasons — the market just happened to go the same direction.
5. **GPT shows weak medium-term holding power** (50% at 4w, 43% at 13w) — marginally better than random, but on only n=8 correct 1d calls. Not enough to be actionable.
6. **HOLD is almost always wrong on volatile stocks.** Both models waste 33-50% of their trades on HOLD, which is correct 0-20% of the time on AMBA. Future work should either eliminate HOLD or set much higher thresholds.
7. **LLM non-determinism is a fundamental barrier.** Sonnet gives different decisions 50% of the time on identical data. Any "alpha" is inseparable from run-to-run randomness.

Recommendations (Updated Post Tri-Monthly Analysis)

1. **Do not deploy the current V5 single-agent system for live trading** — at n=30, no model shows statistically significant edge over Always-BUY at any horizon
2. **The 10-point end-of-month benchmark was misleading** — future evaluations must use at least 30+ data points per model, ideally across multiple tickers, to distinguish signal from noise
3. **Eliminate HOLD on high-volatility stocks** — AMBA's daily moves exceed HOLD thresholds 80-90% of the time; forcing BUY/SELL would at least remove the structural drag
4. **Investigate determinism** — Sonnet's 50% decision consistency on identical data is a fundamental problem; consider temperature=0, structured outputs, or seed parameters to reduce randomness
5. **GPT's medium-term directional accuracy (55-60% on 1w-4w, directional trades only)** warrants further investigation — filter out HOLD trades and test on a wider ticker set

6. **Consider multi-ticker benchmarks (V6)** — the V3 result (440 tasks, 10 tickers, no alpha) already suggested this; the tri-monthly analysis confirms it for AMBA specifically
7. **Focus on the prompt, not the model** — both Sonnet and GPT show the same fundamental inability to predict next-day moves; the issue is likely the input data quality or decision framework, not model capability
8. **Test with real-time indicators** — the current pipeline uses end-of-day data; intraday signals (volume profile, order flow) might provide genuinely predictive features

Limitations

- **Single ticker (AMBA)** — a volatile semiconductor stock; performance could differ on stable large-caps or defensive sectors
- **V4 was a smoke test only** — 10 tickers, 1 date, Haiku only; no formal V4 benchmark with Sonnet/GPT exists
- **V5 Haiku and V5 Sonnet/GPT were separate runs** — same data (cache hit), but different run IDs
- **Cost estimates are approximate** — token usage was not tracked; costs based on per-task model averages
- **No multi-agent comparison in V5** — V2 showed multi-agent has 100% HOLD bias; not re-tested with V5 prompt
- **Walk-forward P2 has only 4 data points** — Sonnet's 75% P2 accuracy (3/4) was likely noise (confirmed by tri-monthly analysis)
- **Tri-monthly analysis confirms n=10 was insufficient** — the 30-point expansion reversed the primary conclusion from "alpha found" to "no alpha"
- **LLM non-determinism** — Sonnet produces different decisions 50% of the time on identical inputs; any single run is a coin flip

Run Inventory

Run ID	Date	Models	Period	Tasks	Completed
v2_20260211_021106	2026-02-11	4 models (V3)	Mar 2025-Jan 2026	440	440
v2_20260214_042556	2026-02-14	Haiku 4.5 (V5)	Mar-Dec 2025	10	10
v2_20260214_132347	2026-02-14	Sonnet 4.5 + GPT 5.1-codex-mini (V5 EoM)	Mar-Dec 2025	20	20
v2_20260214_193006	2026-02-14	Sonnet 4.5 + GPT 5.1-codex-mini (V5 tri-monthly)	Mar-Dec 2025	60	60

Run Files

Run	Checkpoint	Outcomes	Report
V3 (v2_20260211_021106)	checkpoint	outcomes	V3 Benchmark

Run	Checkpoint	Outcomes	Report
V5 Haiku (v2_20260214_042556)	checkpoint	outcomes	—
V5 EoM (v2_20260214_132347)	checkpoint	outcomes	—
V5 Tri-Monthly (v2_20260214_193006)	checkpoint	outcomes	V5 Tri-Monthly Report

Prior Benchmark References

Report	Date	Scope
V2 Comprehensive	2026-02-02	AMBA Jan-Apr 2025, Opus vs OpenAI, single vs multi-agent
V3 10-Ticker	2026-02-11	10 tickers, 4 models, 440 tasks

Resume / Extend Commands

```
# Run V5 Sonnet on 10 tickers (V6 full benchmark)
python -m cli.evaluate_v2 \
    --tickers AAPL,MSFT,NVDA,AMBA,JNJ,XOM,JPM,COST,TSLA,NEE \
    --start 2025-03-01 --end 2025-12-31 \
    --strategies single_sonnet-4-5 \
    --multi-horizon --wait-on-rate-limit

# Compare V5 Sonnet vs V3 Sonnet
python -m cli.evaluate_v2 --compare v2_20260214_132347 v2_20260211_021106
```

Generated 2026-02-14, updated with tri-monthly analysis (v2_20260214_193006). All P&L figures are simulated with \$10,000 portfolio, \$15 commission per trade. V5 prompt includes SPY market context, sector ETF context (AMBA→SMH), and systematic/idiosyncratic risk decision framework. Total evaluation: 90 tasks across 3 runs (10 Haiku EoM + 20 Sonnet/GPT EoM + 60 Sonnet/GPT tri-monthly).